EE3-01
EE9-AC1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2014-15

MSc and EEE PART III/IV: MEng, BEng.and ACGI

ANALOGUE INTEGRATED CIRCUITS AND SYSTEMS

Wednesday, 16 December 9:00 am

Time allowed: 3:00 hours

Corrected copy

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Georgiou
Second Marker(s) : C. Papavassiliou

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1. (a) Figures 1.1 and 1.2 show two constant current generators. Explain the operation of the circuit of Figure 1.1 and state what the key differences between it and the circuit in Figure 1.2. Show equations where necessary.

[6]

(b) Calculate the fractional temperature coefficient for the constant current generators shown Figure 1.1 and 1.2 at room temperature, given that R is a polysilicon resistor with a temperature coefficient of 1500 ppm/°C, the temperature coefficient of V_{BE} to be $-2.5mV/°C$ both at T=300K. State all assumptions.

[6]

(c) Voltage and current-sources are key components in analogue circuit design. Sketch a typical band-gap voltage reference circuit and prove that the temperature coefficient of the output voltage V_0 is zero if $V_0 = 1.283$ V. Assume the temperature coefficient of V_{BE} to be $-2.5mV/°C$, Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K and electron charge $q = 1.6 \times 10^{-19}$ C.

[8]

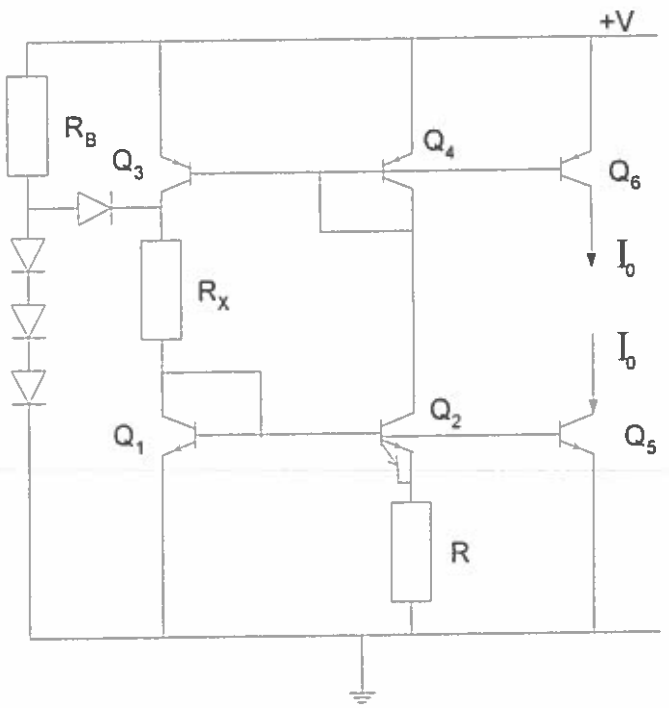


Figure 1.1

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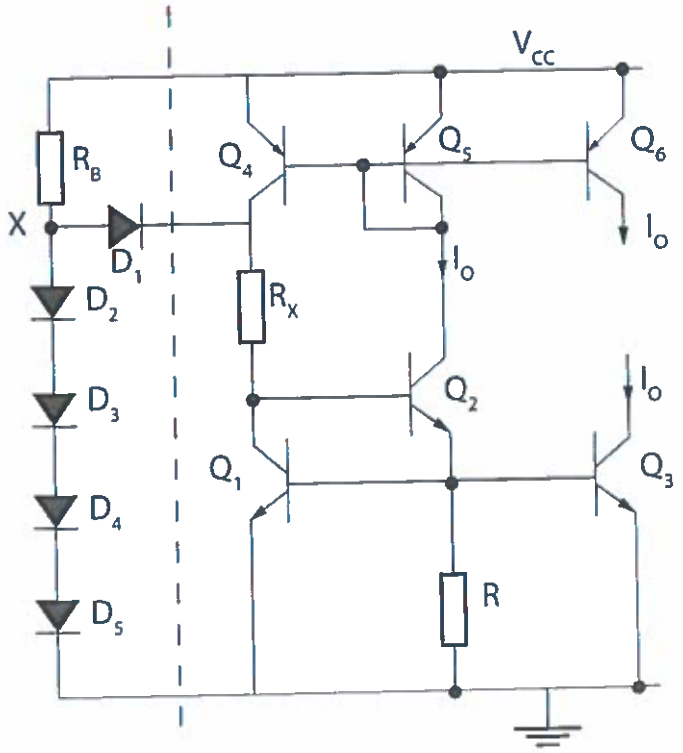


Figure 1.2

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2. (a) Give one advantage and one disadvantage of each of the two CMOS current mirror circuits shown in Figures 2.1 and 2.2.

[4]

(b) For the current mirror of Figure 2.2 derive this voltage swing in terms of device threshold voltage V_T, clearly stating any assumptions you make.

[6]

(c) With the aid of a macromodel, explain the theoretical concept of current feedback and how it results in constant bandwidth application.

[10]

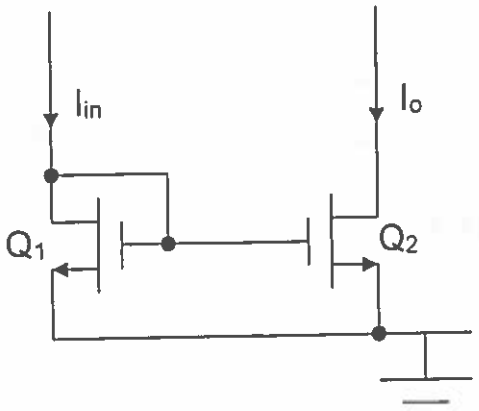


Figure 2.1

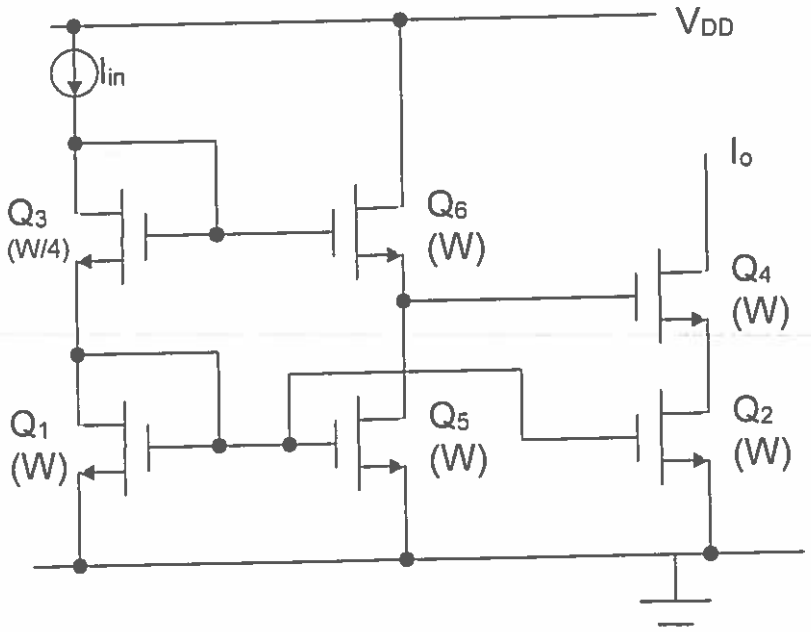


Figure 2.2

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3. (a). Figure 3 shows a two-stage CMOS op-amp.

(i) Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below.

[8]

(ii) What is the main advantage and disadvantage of using a single-stage op-amp instead of a two-stage op-amp ?

[2]

(b).

(i) Show how the output transistors can be cascoded to increase the voltage gain.

[6]

(ii) State two advantages and two disadvantages of using a cascoded output stage.

[4]

CMOS TRANSISTOR PARAMETERS

MODEL PARAMETERS	Kp ($\mu A/V^2$)	λ (V^{-1})	V_{TO} (V)
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

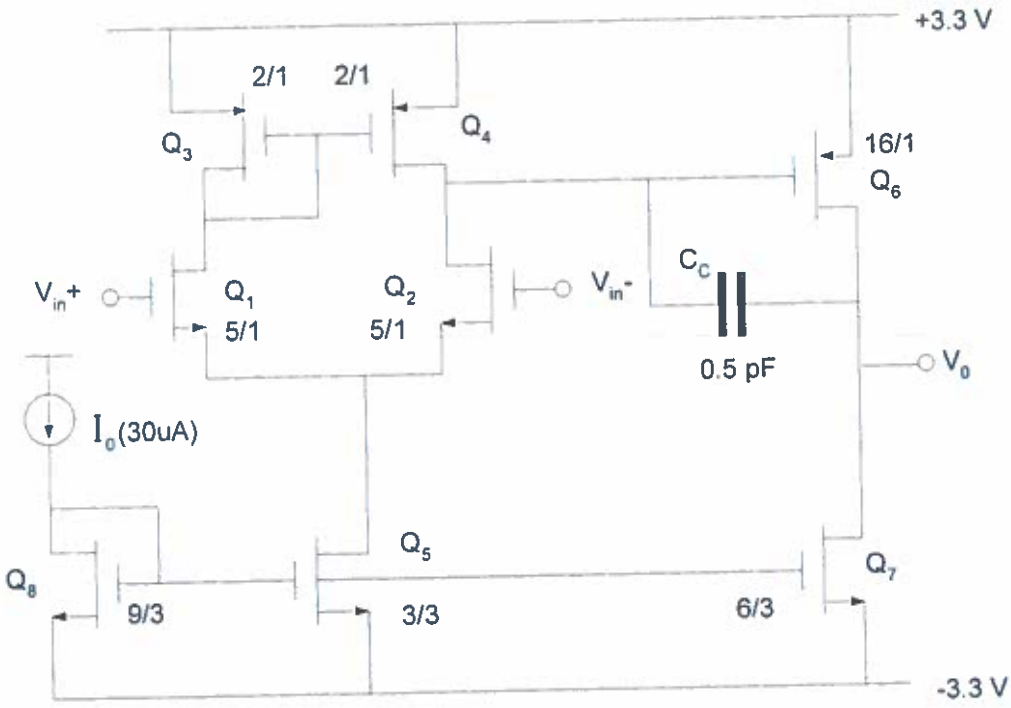


Figure 3

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4. Figure 4 shows a differential opamp which is based upon a folded cascode design

(a). Explain what is meant by folded cascode.

[4]

(b). Sketch a circuit diagram of the folded cascode opamp including a circuit which ensures the common mode voltage is maintained at the output.

[6]

(c). (i) Derive the equation for the linear differential resistor of the circuit shown in Figure 4.

[5]

(ii) Assuming the time constant of the circuit is τ=10ms, calculate the value of feedback capacitor, C, assuming input transistors Q1 and Q2 are operating in the triode region and have W/L=5, K=20uA/V2, Vg=1V, Vt=0.5V and node x is at a common mode of 0 V. You may neglect any body effects.

[5]

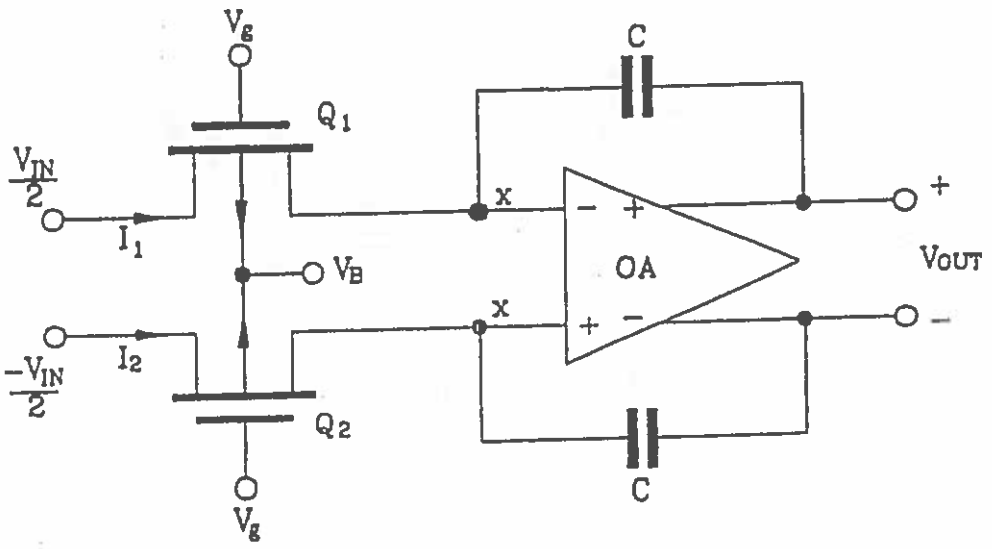


Figure 4.

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5. (a) (i) Explain the principles of sigma delta modulation. [5]

(ii) Sketch a typical circuit architecture of a sigma delta modulator. [5]

(b) Figure 5 shows one bit of an algorithmic data converter.

(i) Explain how the data conversion works. [3]

(ii)Show how this can be cascaded to perform an 8-bit data conversion. [4]

(c) What noise limits the resolution of sampled-data converters? [3]

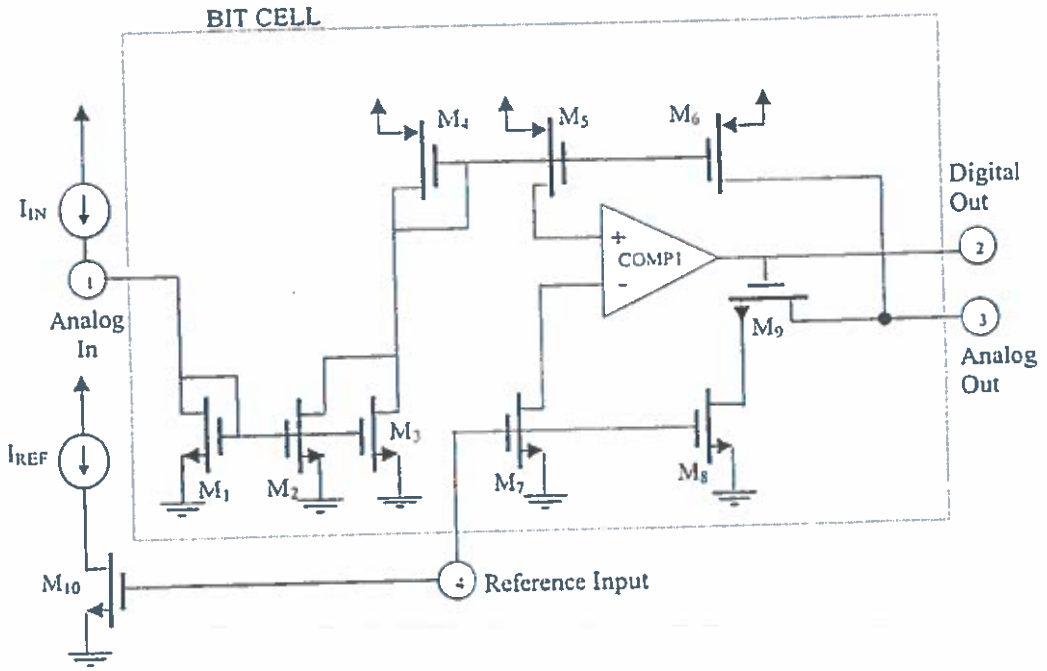


Figure 5.

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6. (a) Sketch the basic design of a 3rd-order Chebyshev low pass switched-capacitor ladder filter using op-amps, capacitors and switches.

[5]

(b) For the circuit in part a. state the equations of the normalised passive component values and draw the equivalent RLC prototype.

[5]

(c) From the circuit, calculate normalised passive component values for the original double terminated LC prototype of the filter. The filter is to have a cut-off frequency of 5kHz and assume a clocking frequency of 100 kHz. The values of the integration capacitor for the capacitor-based sections are 5.06pF, and for the inductive based sections is 3.49pF. All other switched capacitors are 1pF. All values should be normalised to 1rad/s. You may assume that the clocking frequency is so high that the integrators can be assumed lossless.

[10]

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